SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment 6.5

Program	**: B. Tech (CSE)**
Course Title	**: AI Assisted Coding**
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Task 1

Prompt 1: Write a Python code to check voting eligibility based on age and citizenship using condtitional statements

Code:

```python
age=input("Enter your age: ")
citizenship=input("Are you a citizen? (yes/no): ")
if age.isdigit() and int(age) >= 18 and citizenship.lower() == 'yes':
    print("You are eligible to vote.")
else:
    print("You are not eligible to vote.")
```

Output:

```
Enter your age: 20
Are you a citizen? (yes/no): yes
You are eligible to vote.
```

Explanation:

The code first takes the user's age and citizenship status as input.
It then checks if the age is a digit and if it is 18 or older.
Additionally, it checks if the citizenship input is 'yes' .

TASK2

Prompt : Python code to count vowels and consonants in a string using loop

Code:

```python
string = input("Enter a string: ")
vowels = "aeiouAEIOU"
vowel_count = 0
consonant_count = 0

for char in string:
    if char in vowels:
        vowel_count += 1
    elif char.isalpha():
        consonant_count += 1

print(f"Vowels: {vowel_count}, Consonants: {consonant_count}")
```

Output:

```
Enter a string: Prashanth
Vowels: 2, Consonants: 7
```

Explanation:

The code takes a string input from the user and initializes counters for vowels and consonants It then iterates through each character in the string, checking if it is a vowel or consonant, and increments the respective counter. Finally, it prints the counts of vowels and consonants.

TASK 3:

Prompt: Python program for a library management system using classes, loops, and conditional statements.

Code:

```python
from functools import reduce
class Library:
    def __init__(self):
        self.books = []

    def add_book(self, book_name):
        self.books.append(book_name)
        print(f'Book "{book_name}" added to the library.')

    def remove_book(self, book_name):
        if book_name in self.books:
            self.books.remove(book_name)
            print(f'Book "{book_name}" removed from the library.')
        else:
            print(f'Book "{book_name}" not found in the library.')

    def display_books(self):
        if self.books:
            print("Books available in the library:")
            for book in self.books:
                print(f'- {book}')
        else:
            print("No books available in the library.")
def main():
    library = Library()
    while True:
        print("\nLibrary Management System")
        print("1. Add Book")
        print("2. Remove Book")
        print("3. Display Books")
        print("4. Exit")
        choice = input("Enter your choice (1-4): ")

        if choice == '1':
            book_name = input("Enter the name of the book to add: ")
            library.add_book(book_name)
        elif choice == '2':
            book_name = input("Enter the name of the book to remove: ")
            library.remove_book(book_name)
        elif choice == '3':
            library.display_books()
        elif choice == '4':
            print("Exiting the Library Management System.")
            break
        else:
            print("Invalid choice. Please try again.")
if __name__ == "__main__":
    main()
```

Output:

```
1. Add Book
2. Remove Book
2. Remove Book
2. Remove Book
3. Display Books
4. Exit
3. Display Books
4. Exit
Enter your choice (1-4): 4
4. Exit
Enter your choice (1-4): 4
Enter your choice (1-4): 4
Exiting the Library Management System.
```

Explaination:
This Python program implements a simple library management system using classes, loops, and conditional statements.
The Library class contains methods to add, remove, and display books in the library.
The main function provides a user interface to interact with the library system through a menu-driven approach
The program runs in a loop until the user chooses to exit, allowing multiple operations on the library.

TASK 4:

Prompt: Python class to mark and display student attendance using loops.
CODE:

```python
class StudentAttendance:
    def __init__(self):
        self.students = []
        self.attendance_records = {}

    def add_student(self, student_name):
        self.students.append(student_name)
        self.attendance_records[student_name] = []

    def mark_attendance(self, student_name, status):
        if student_name in self.students:
            self.attendance_records[student_name].append(status)
        else:
            print(f"Student {student_name} not found.")

    def display_attendance(self):
        for student in self.students:
            print(f"{student}: {self.attendance_records[student]}")
# Example usage
attendance = StudentAttendance()
attendance.add_student("Bob")
attendance.add_student("Alice")
attendance.mark_attendance("Alice", "Present")
attendance.mark_attendance("Bob", "Absent")
attendance.mark_attendance("Alice", "Present")

attendance.display_attendance()
```

OUTPUT:

```
Bob: ['Absent']
Alice: ['Present', 'Present']
```

Explanation:

The StudentAttendance class allows you to manage student attendance. You can add students, mark their attendance as "Present" or "Absent", and display the attendance records for all students using loops.

TASK5

PROMPT: Python program using loops and conditionals to simulate an ATM menu.

CODE:

```python
def atm_menu():
    balance = 1000  # Initial balance
    while True:
        print("\nWelcome to the ATM")
        print("1. Check Balance")
        print("2. Deposit Money")
        print("3. Withdraw Money")
        print("4. Exit")

        choice = input("Please select an option (1-4): ")

        if choice == '1':
            print(f"Your current balance is: ${balance}")

        elif choice == '2':
            deposit = float(input("Enter amount to deposit: $"))
            if deposit > 0:
                balance += deposit
                print(f"${deposit} deposited successfully.")
            else:
                print("Invalid amount. Please enter a positive number.")

        elif choice == '3':
            withdraw = float(input("Enter amount to withdraw: $"))
            if 0 < withdraw <= balance:
                balance -= withdraw
                print(f"${withdraw} withdrawn successfully.")
            else:
                print("Invalid amount. Please check your balance and try again.")

        elif choice == '4':
            print("Thank you for using the ATM. Goodbye!")
            break

        else:
            print("Invalid selection. Please choose a valid option.")
# Run the ATM menu
atm_menu()
```

OUTPUT:

```
1. Check Balance
2. Deposit Money
3. Withdraw Money
2. Deposit Money
3. Withdraw Money
3. Withdraw Money
4. Exit
4. Exit
4. Exit
Please select an option (1-4): 
```

Explanation:

This program simulates a simple ATM menu using a while loop to continuously display options until the
user decides to exit. It uses conditionals to handle different user choices such as checking balance,depositing money, withdrawing money, and exiting program. The balance is updated accordingly based on user actions. on deposits and withdrawals.